

Zero-Shot Macaque Face Detection with SAM 3: A Prompt-Benchmark and Comparison Against a Trained YOLO Detector

FacialRecognitionTest project — sam3_experiments/ (2026-05-27)

Abstract

We evaluate the zero-shot, text-promptable Segment Anything Model 3 (SAM 3) on the task of macaque face detection in field photographs of the Gibraltar Barbary macaque (*Macaca sylvanus*) population, and compare it against an in-house YOLOv8 face detector trained on the same dataset. Two complementary experiments were run on an NVIDIA A100 80 GB GPU: (i) a quantitative prompt benchmark against ground-truth bounding boxes on a 50-image stratified sample of the YOLO validation split, sweeping 13 candidate text prompts under a common confidence threshold (score > 0.05) and non-maximum suppression pipeline (IoU > 0.5); and (ii) a qualitative three-scenario test (single face, multi-face, macaque-with-human) on real Gibraltar and Instagram tourist photographs. The trained YOLO detector achieved AP50 = 0.981 on the full validation set (n = 1,333; 0.979 on the 50-image sample), with precision = recall = F1 = 0.98. The best SAM 3 prompt, “monkey face”, reached AP50 = 0.621 (precision 0.57, recall 0.78, F1 = 0.66), 0.358 AP50 below the trained detector. Counter-intuitively, species-specific prompts (“macaque face”, “Barbary macaque face”) underperformed the generic taxon-level prompt, consistent with the relative frequencies of these terms in CLIP-style pre-training corpora. In Scenario 3 (macaque + human), SAM 3 recovered human faces that the macaque-specific YOLO model is incapable of detecting in 4 / 10 images, demonstrating a complementary role for SAM 3 as an open-vocabulary detector. We conclude that a trained YOLO model remains the operational detector of choice for the existing macaque re-identification pipeline, while SAM 3 is best deployed for open-vocabulary tasks (human-presence filtering, novel-prompt exploration) where labelled data are unavailable.

1. Introduction

Reliable face localisation is the first stage of any individual re-identification pipeline. In the FacialRecognitionTest project we have built and trained a YOLOv8 macaque-face detector that powers the subsequent embedding and identification stages. Meta’s release of Segment Anything Model 3 (SAM 3), which accepts a free-form text prompt and returns segmentation masks plus bounding boxes, raises the natural question of whether a general-purpose, zero-shot model can replace or supplement the trained detector. SAM 3 also opens the door to operations that are out of scope for a species-specific YOLO model, most notably the detection of human faces in the same frame as macaques — an important downstream consideration for tourist-photo data sources (Instagram, Cable Car location).

This report summarises the experiments conducted in sam3_experiments/ during February 2026. We address three questions:

- (Q1) Which textual prompt elicits the strongest macaque-face detector from SAM 3?
- (Q2) How does the best SAM 3 prompt compare quantitatively to the trained YOLO baseline?
- (Q3) Does SAM 3 expand the operational envelope of the existing pipeline (multi-face frames, mixed macaque/human scenes)?

2. Materials and Methods

2.1 Hardware and software

All experiments were run on a single NVIDIA A100-SXM4-80 GB GPU (node gpu-q-2 / gpu-q-42), with PyTorch 2.7.0 + CUDA 12.6, Python 3.12 inside a dedicated conda environment (sam3). The SAM 3 model weights were obtained from the facebook/sam3 HuggingFace repository (gated access). SAM 3 does not currently support CPU inference; all code paths assume CUDA. The YOLO baseline used the in-house checkpoint macaque_face_detector_20260120_v1/best.pt loaded via ultralytics 8.4.14.

2.2 Datasets

Two image sources were used. (i) The Gibraltar Macaques dataset comprises approximately 9,900 field photographs organised by location and named individual (folders females/ and males/). For the ground-truth benchmark, the existing YOLO validation split was used (1,333 images, one face per image as the dominant case; mean face count 1.0). A seeded sample of 50 images was drawn from this split for the prompt sweep. (ii) The IG_Photos dataset ($n = 10$) contains real Instagram-scraped tourist photographs that include both human and macaque faces in the same frame, and is used exclusively for the Scenario 3 qualitative test.

2.3 Model invocation and post-processing

Each input image was passed to SAM 3 in image-mode (Sam3Processor) with a single text prompt per inference call. The raw output (boxes, scores, masks) was passed through a two-step filter:

- (a) confidence threshold: boxes with score ≤ 0.05 were discarded;
- (b) non-maximum suppression at IoU > 0.5 (with an additional intersection-over-min-area rule of 0.5 to catch the case where a tight box is nested inside a loose box around the same face).

These same thresholds were applied identically to every prompt so the comparison is fair. For YOLO, the standard confidence threshold of 0.25 was used (Ultralytics default).

2.4 Prompts evaluated

Thirteen prompts were swept, spanning four families: species-specific (“macaque face”, “Barbary macaque face”, “Barbary macaque”), taxon-level (“monkey face”, “monkey head”, “macaque head”, “primate face”), visual/descriptive (“furry animal face”, “close-up of a monkey face”, “animal face”), body-level (“macaque”, “monkey”), and a zero-shot ceiling (“face”).

2.5 Evaluation metrics

Predicted boxes were matched to ground-truth boxes by greedy highest-IoU assignment; a prediction was a true positive if its best IoU with an unmatched GT box was ≥ 0.5 . We report precision = $TP / (TP + FP)$, recall = $TP / (TP + FN)$, F1, and AP50 (area under the precision–recall curve at IoU 0.5). Per-image AP50 was also recorded; the dataset-level AP50 reported here is the mean of per-image values as written by benchmark_prompts.py.

3. Experiments

3.1 Quantitative prompt benchmark (Experiment A)

Each of the 13 prompts was applied to every image in the 50-image validation sample; predicted boxes were filtered and NMS-deduplicated, and matched against the YOLO-format ground truth labels at $\text{IoU} \geq 0.5$. Per-image side-by-side visualisations (GT vs. SAM 3 vs. YOLO predictions) are saved under `results/eval_vs_yolo/` together with the summary CSV. For reference, the trained YOLO model was first evaluated on the full validation set ($n = 1,333$) using its own `val()` routine to anchor the numbers reported on the 50-image sample.

3.2 Qualitative scenario test (Experiment B)

A complementary three-scenario test was run on raw photographs to characterise SAM 3’s behaviour outside the labelled benchmark.

S1 — Single macaque face: three images sampled (seed = 42) from individual folders, with no “+” character in the filename (single named animal).

S2 — Multiple macaque faces: three images whose filenames contain “+”, indicating two or more named animals (23 such images exist in the corpus).

S3 — Macaque + human face: all 10 IG_Photos images; for each frame, SAM 3 was run twice (prompts “monkey face” and “human face”) and the union plotted in two colours.

Throughout Scenario 3 the SAM 3 prompt “monkey face” was used as it was the top-ranked prompt from Experiment A.

4. Results

4.1 YOLO baseline

On the full validation set (1,333 images, 1,333 instances), the trained YOLO detector achieved $\text{AP}_{50} = 0.981$ and $\text{AP}_{50-95} = 0.600$, with $P = 0.976$, $R = 0.964$ (Ultralytics `val()`). On the 50-image sample, the matched metrics were $\text{AP}_{50} = 0.979$, $P = 0.98$, $R = 0.98$, $F1 = 0.98$, $\text{TP} = 49$, $\text{FP} = 1$, $\text{FN} = 1$, confirming that the sample is representative of the full split (sample AP_{50} within 0.002 of full).

4.2 Prompt benchmark

Table 1 reports the 13 prompts ranked by AP_{50} , with the YOLO reference row at the top. Counts are TP/FP/FN against 50 ground-truth boxes (one per image).

Table 1. SAM 3 prompt benchmark on the 50-image validation sample ($\text{IoU} \geq 0.5$). YOLO (trained) is the in-house macaque face detector baseline. Bold = best SAM 3 prompt.

Rank	Detector / Prompt	AP50	Precision	Recall	F1	TP	FP	FN
REF	YOLO (trained)	0.979	0.980	0.980	0.980	49	1	1
1	SAM3: monkey face	0.621	0.565	0.780	0.655	39	30	11

2	SAM3: monkey head	0.585	0.708	0.680	0.694	34	14	16
3	SAM3: macaque head	0.477	0.690	0.580	0.630	29	13	21
4	SAM3: face	0.452	0.589	0.660	0.623	33	23	17
5	SAM3: close-up of a monkey face	0.422	0.397	0.620	0.484	31	47	19
6	SAM3: macaque face	0.346	0.587	0.540	0.563	27	19	23
7	SAM3: primate face	0.269	0.682	0.300	0.417	15	7	35
8	SAM3: Barbary macaque face	0.238	0.700	0.280	0.400	14	6	36
9	SAM3: furry animal face	0.063	0.571	0.080	0.140	4	3	46
10	SAM3: animal face	0.040	1.000	0.040	0.077	2	0	48
11	SAM3: monkey	0.004	0.051	0.060	0.055	3	56	47
12	SAM3: Barbary macaque	0.003	0.038	0.040	0.039	2	50	48
13	SAM3: macaque	0.001	0.026	0.020	0.023	1	37	49

Three findings stand out. First, the best SAM 3 prompt (“monkey face”) trails the trained YOLO detector by 0.358 AP50 (0.621 vs. 0.979) and by 0.32 F1 (0.66 vs. 0.98) on the same images. Second, taxon-level prompts (“monkey *”, AP50 0.585–0.621) consistently outperform species-specific prompts (“macaque face” 0.346; “Barbary macaque face” 0.238). Third, dropping the “face” / “head” qualifier collapses performance: “monkey” alone scores AP50 = 0.004 (FP = 56) because SAM 3 then returns whole-body boxes that fail the IoU = 0.5 test against face-only GT. The bias is therefore concentrated at two axes: lexical frequency in pre-training data (monkey $\gg$ Barbary macaque) and the presence of an explicit anatomical anchor (“face”/“head”).

4.3 Scenario test

Table 2 reports the per-image detection counts in the three qualitative scenarios, using the best prompt “monkey face” (and additionally “human face” in S3).

Table 2. Per-image detection counts for the qualitative scenarios. S3 reports SAM 3 results decomposed by prompt; the “SAM 3 total” column is the union over both prompts after independent NMS per prompt.

Scenario	Image	YOLO	SAM 3 (monkey face)	SAM 3 (human face)	SAM 3 total
S1	Abby (Apes Den)	2	0	—	0
S1	Jordi (Prince Philip Arch)	1	1	—	1
S1	Jamie temporary (O’Hara)	1	1	—	1
S2	Tripatte + AF2 (Rock Gun)	1	0	—	0
S2	Flick + follow (O’Hara)	1	1	—	1
S2	Alexa + SAM (O’Hara)	2	2	—	2
S3	image_10	1	2	1	3
S3	image_4	2	2	0	2
S3	image_8	1	2	1	3
S3	image_44	1	1	1	2
S3	image_33	2	0	1	1
S3	image_5	1	1	0	1
S3	image_28	1	1	1	2
S3	image_2	2	2	0	2
S3	image_18	2	2	0	2
S3	image_3	1	1	0	1

Aggregated over the 10 IG_Photos frames in S3, YOLO returned 14 macaque detections, SAM 3 returned 14 macaque detections plus 5 human-face detections that YOLO cannot, by construction, produce. SAM 3 exceeded YOLO’s box count in 4 of 10 S3 frames (image_10, image_8, image_44, image_28) and was at parity on 5; on image_33 SAM 3 missed both macaques while still recovering a human face. In S1/S2, SAM 3 matched YOLO on 4 of 6 frames and missed faces present in YOLO output on 2 (Abby, Tripatte) — both are field photographs in difficult lighting and partial occlusion conditions.

5. Discussion

5.1 Why generic prompts beat species-specific ones

The 0.275 AP50 gap between “monkey face” and “macaque face”, and the 0.383 gap between “monkey face” and “Barbary macaque face”, are too large to attribute to noise on a 50-image sample. The most plausible explanation is the underlying distribution of these terms in the image–text corpora used to pre-train SAM 3’s text encoder: web captions contain orders of magnitude more occurrences of “monkey” than of “macaque”, let alone “Barbary macaque”. The encoder therefore produces a stronger and better-calibrated visual concept for the common term. This is consistent with prior reports on CLIP-style encoders, where rare-class taxonomic names lag behind common-language labels.

5.2 Anatomical anchors matter

Removing the word “face” or “head” from a prompt drops AP50 by two orders of magnitude (0.621 → 0.004 for “monkey face” → “monkey”). SAM 3 then returns whole-animal boxes that do not satisfy the $\text{IoU} \geq 0.5$ condition against face-only ground truth. For pipelines downstream of detection (face embedding, individual re-identification) this matters: a body-box is not interchangeable with a face-box. Any SAM 3 deployment for face detection should include an explicit anatomical anchor in the prompt.

5.3 Where SAM 3 adds value over the trained detector

The Scenario 3 results expose a capability the trained YOLO model does not have. The macaque face detector was trained on macaque faces only and cannot, by design, flag human faces in the same frame. In the IG_Photos dataset, where tourist selfies routinely include both humans and macaques, SAM 3’s zero-shot “human face” prompt recovered five human-face detections across the ten frames at the same confidence and NMS thresholds. This is directly useful: it offers a drop-in pre-filter to either exclude human-containing frames from the macaque re-identification pipeline or to flag them for manual review, without retraining a multi-class detector.

5.4 Limitations

Three limitations should be flagged. (i) Sample size: the 50-image benchmark is a representative sample (YOLO AP50 within 0.002 of the full split) but a 200–500 image evaluation would tighten the prompt ranking, especially in the middle of the table where AP50 differences are within 0.05–0.10. (ii) Single-prompt inference: each SAM 3 call uses one prompt; an ensemble of the top 2–3 prompts followed by cross-prompt NMS was not tested and is likely to improve recall. (iii) Confidence threshold: a fixed score > 0.05 was used to keep the prompt comparison fair, but for downstream use the threshold should be re-tuned per prompt against an operating-point preference (precision vs. recall).

6. Conclusion

On macaque face detection in the FacialRecognitionTest corpus, a trained YOLO detector remains substantially ahead of zero-shot SAM 3 (AP50 0.979 vs. 0.621; F1 0.98 vs. 0.66). However, SAM 3 is not redundant: it provides a low-effort, open-vocabulary tool for tasks the macaque-specific YOLO cannot address — most importantly the detection of human faces in mixed scenes. We recommend keeping the trained YOLO detector as the primary face localiser in the re-identification pipeline, and using SAM 3 with the prompt “monkey face” (or an ensemble with “monkey head”) as an auxiliary detector for (a) frames containing human bystanders, (b) prompts beyond the YOLO training classes, and (c) data-curation passes where labelled data are not yet available.

Appendix A. Files and reproducibility

All scripts and outputs live under `sam3_experiments/`.

- `benchmark_prompts.py` — prompt sweep and ground-truth evaluation (`--eval mode`)
- `test_sam3_macaque.py` — three-scenario qualitative test
- `run_sam3_gpu.sh` — SLURM submission wrapper (A100, conda env `sam3`)
- `results/eval_vs_yolo/eval_vs_yolo.csv` — prompt benchmark results
- `results/eval_vs_yolo/eval_*.jpg` — 50 side-by-side comparisons (GT, YOLO, SAM 3 best)

- results/scenario_20260225_044910/ — final scenario run, 16 panels ($S1 \times 3$, $S2 \times 3$, $S3 \times 10$)
- logs/23352722.out — full eval log
- logs/23434961.out — final scenario log

Random seed = 42 throughout. SAM 3 inference is deterministic at the image level given a fixed prompt and post-processing pipeline.